

A330

VER. 25.1.23

KneeBoard



Click for Update

by Flyingdeuk

Domestic

Japan

China

S.E Asia(GUM)

Supplement

FUEL Consumption

LP Aircon/ HP StartUnit

Cold Temp Correction

Cold Wx Operation

ENG ON Deicing

ENG OFF Deicing

Differences (ATS A333)

A322(5) HL7524-7551	APMS	EY 252 (151) EY 260 (156)	Total PAX 200 NOR CHK OFF PAX No HL 8026, 8027 PAX AUTO
A323(11) HL7553-7720			
A223E(3) HL82xx	APMS -2.5	EY 188 (112)	
A323E(6) HL80xx	APMS -1.5	EY 248 (148) HL8001 RNP AR X	
ENG START		HL7xxx 43-48%	HL8xxx 52-54%

PRINCIPAL DIMENSION

WINGSPAN : ICAO E, FAA V
Weight CAT : Heavy

	A330-300 / 300E		A330-200 / 200E	
LENGTH	63.69 m	208.92 ft	58.37 m	191.25 ft
WING SPAN	60.30 m	197.83 ft	60.30 m	197.83 ft
HEIGHT	16.83 m	55.25 ft	17.80 m / 17.30 m	58.42 ft / 56.67 ft
WHEEL BASE	25.37 m	83.25 ft	22.18 m	72.75 ft
WHEEL TREAD	10.68 m	35.08 ft	10.68 m	35.08 ft

Refer to FCOM DSC-20-20 Principal Dimensions **180 Back 72도 48m/(157ft)**

MTOW

238T	524,700 lbs	230T	507,063 lbs
217T	468,402 lbs	198T	436,515 lbs
192T	423,287 lbs		

Domestic

GMP

CJU

GMP

PUS

CJU

PUS

ICN

PUS

Welcome PA

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WELCOME PA

손님 여러분, 안녕하세요?

저는 여러분을 모시고 가는 기장 ____입니다.

저희 대한항공을 이용해 주셔서 대단히 고맙습니다.

____(국제)공항까지 비행시간은 ____시간 ____분
으로 예상됩니다.

비행 중에는 항공기가 갑자기 흔들릴 수도 있으니,
자리에 앉아 계실 때에는 항상 좌석벨트를
매주시기 바랍니다. (저희 승무원 모두는)
저는 여러분을 안전하게 모시기 위해 최선을
다하겠습니다. 고맙습니다.

Good morning (afternoon /evening), ladies and
gentlemen.

This is captain last name speaking.

Welcome aboard Korean Air.

This flight is bound for ____ (international)
airport and our flight time is ____ hours(s) and
minutes.

For your safety, keep your seatbelts fastened
while you are seated.

Thank you for choosing Koreanair.

Please enjoy your flight.

Domestic

GMP

서울/김포국제

ICN

서울/인천국제

CJU

제주국제

PUS

부산/김해국제

도착 방송 (5시간이상, 40분전)

출발지 기준 2200-0800 Quiet Hour

손님 여러분, 저는 기장입니다.

우리 비행기는 앞으로 약 (40)분 후에

___국제공항에 착륙 예정입니다.

현재 공항의 날씨는 ①___, 기온은 섭씨 ___도 입니다.

① 맑으며

① (다소)흐리며

① (이슬)비가 내리며/소나기가 내리며

① 바람이 불고 있으며

① 눈이 오고 있으며

① 안개가 끼어 있으며

① 황사가 있으며

지금 이곳의 시각은 __월 __일 __요일, 오전(오후)

__시 __분 입니다.

고맙습니다.

Ladies and gentlemen, this is the captain speaking.

We expect to land at ___ international airport in about
(40) minutes.

The current temperature at ___ is ___ degrees Celsius,
or ___ degrees Fahrenheit (NAVBLUE 참조)

and it is ①___.

① (mostly) clear

① (partly) cloudy

① drizzling / raining

① windy

① snowing

① foggy

① hazy or smoggy

The current time is __ : __ a.m(p.m), on (day-of-the-
week), (month)(date).

Thank you for flying with us today.

Domestic

Japan

[GMP](#)

[KIX](#)

[ICN](#)

[KIX](#)

[ICN](#)

[NRT](#)

[ICN](#)

[CTS](#)

[ICN](#)

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Please enjoy your flight.

Japan	
KIX	오사카/간사이
HND	도쿄/하네다
NRT	도쿄/나리타
CTS	삿포로/신(New) 치토세
NGO	나고야/주부(Chubu Centrair)
FUK	후쿠오카

Japan

China

[GMP](#)

[SHA](#)

[GMP](#)

[PEK](#)

[ICN](#)

[TAO](#)

[ICN](#)

[PEK](#)

[ICN](#)

[SHE](#)

[ICN](#)

[PVG](#)

[ICN](#)

[HKG](#)

[ICN](#)

[DLC](#)

[ICN](#)

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China

SHA

상하이/홍차오

TAO

칭다오/자오둥

PEK

베이징/소우뚜(Capital)

SHE

선양/타오셴

PVG

상하이/푸둥

HKG

홍콩

TSN

톈진/빈하이

DLC

다렌/쵸우쑤이쯔

KMG

쿤밍/창쉐이

CHINA

S.E Asia

ICN

CXR

ICN

SGN

ICN

PNH

ICN

MNL

ICN

RMQ

ICN

TPE

PUS

TPE

ICN

GUM

PUS

BKK

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Please enjoy your flight.

S.E Asia

CXR 베트남 나짱(Nha Trang)/깜라인

SGN 베트남 호찌민/탄소넛

PNH 캄보디아 프놈펜

MNL 필리핀 마닐라/니노이 아키노

TPE 타이완/타이페이 타오유엔

RMQ 타이완/타이중 칭찬강

PGUM 괌

BKK 방콕/수완나폼

도착 방송 (5시간이상, 40분전)

출발지 기준 2200-0800 Quiet Hour

손님 여러분, 저는 기장입니다.

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① 맑으며

① (다소)흐리며

① (이슬)비가 내리며/소나기가 내리며

① 바람이 불고 있으며

① 눈이 오고 있으며

① 안개가 끼어 있으며

① 황사가 있으며

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고맙습니다.

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(40) minutes.

The current temperature at __ is __ degrees Celsius,
or __ degrees Fahrenheit (NAVBLUE 참고)

and it is ① __.

① (mostly) clear

① (partly) cloudy

① drizzling / raining

① windy

① snowing

① foggy

① hazy or smoggy

The current time is __ : __ a.m(p.m), on (day-of-the-
week), (month)(date).

Thank you for flying with us today.

RKSS(GMP) 59ft

RKPC(CJU) 119ft

KE GMP 131.15

DCL -15분 가능 TOBT 5분 차이

시 CTC Comm

PA

KE CJU 129.4



Rwy 32R **Takeoff**

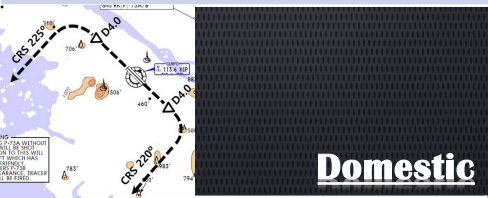
(06:00L~0900L / 12:00L~15:00L

/18:00L~21:00L)

GMP : **NADP FX-10도, CLB, CONF 1+F, Acc 4000ft**

32L/R	BULTI xT	HD 324 ALT 5000 Mod NADP		
	(BULTI xQ)			
14L/R	BULTI xU	HD 144 ALT 6000 Mod NADP		
	(BULTI xZ)			
KIP 113.6	32L 108.3	32R 110.7	14L 109.9	14R 108.7
FIX	32L/R : EO32L/R, R225, YJU R271		14L/R : EO14L/R, R220, P73 /2	

APRON(130.875) -> GND(121.9) -> TWR (All by ATC)



CJU : STAR

AFT Merge PT(220kts) DCT IAF(210kts), FAF (160kts)

ILS Z 07	DOTOL xP	YUMIN	DOTOL 160
RNP Y 07 (No LPV), RNP Z 07 AR (RNP 0.11)			
ILS Z 25	DOTOL xT(xM)	DUKAL	DOTOL/-10 160
FIX	RWxx /8		

07 : P6(5176'), P5(5882'), P4(6840'-ATC HIRO)

25 : P7(5219'), P8(5882'), P10(7524'-ATC HIRO)

Entering Rapid TWY CTC GND 121.675 (STOP x)

HST 40KTS

RKPC(CJU) 119ft

RKSS(GMP) 59ft

KE CJU 129.4

DCL -10분

PA

KE GMP 131.15

Rwy 32L **Landing**

(06:00L~0900L / 12:00L~15:00L
/18:00L~21:00L)



CJU : SID (NADP 1)

07

KAMIT xE

HD 066 ALT 10000

25

KAMIT xW

HD 246 ALT 10000

YDM 109.0

07 109.9

25 111.3

HUD

07 : NONE

25 : YDM246/3, R290

07 : Passing G4 CTC TWR

25 : 31 Holding PSN on P, E1,2,3 CTC TWR



Domestic

GMP : STAR

ILS 32L/R

OLMEN xT

BUMSI

OLMEN 160

ILS 14R

OLMEN xU

DOKDO

OLMEN 160

FIX

KIP /8(RWY 32), YJU R271, P73 /2

32L : D3(6532'), E2(9117'), 32R(**No CL L/T**) : E1(6614')
14R : C1(6578')

32L/R : 8 KIP L/G, 14R : LOC CAPT L/G

FAF : Final Flap

TWR -> GND -> APRON (All by ATC)

Except RWY14R Landing (Until R)

RKSS(GMP) 59ft | RKPK(PUS) 13ft

KE GMP 131.15 **PA** KE Gimhae 129.2
 DCL -15분 가능 TOBT 5분 차이
 시 CTC Comm

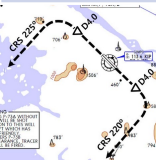


Rwy 32R **Takeoff**
 (06:00L~0900L / 12:00L~15:00L
 /18:00L~21:00L)

GMP : NADP FX-10도, CLB, CONF 1+F, Acc 4000ft

32L/R	OSPOT xT	HD 324 ALT 5000 Modi NADP		
	(OSPOT xQ)			
14L/R	OSPOT xU	HD 144 ALT 6000 Modi NADP		
	(OSPOT xZ)			
KIP 113.6	32L 108.3	32R 110.7	14L 109.9	14R 108.7
HUD	32L/R : EO32L/R, R225, YJU R271		14L/R : EO14L/R, R220, P73 /2	

APRON(130.875) -> GND(121.9) -> TWR (All by ATC)



Domestic

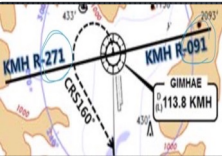
PUS : STAR (T/W 10tks, 36R, 360000lbs 제한)

ILS 36	KEVOX x	MASTA	9DME LG, 8DME FLAP
VOR 18	GAYHA x	MASTA	<u>18 Circling Click!!</u>
FIX	36 : IKMA/IKHE /9, /8		18 : KMH R284, R280

36L : C4 (6299'), C2(7795') 36R : E3(5866'), E2(7339')
 18R(DIS):C6(5770'),C7(6824') 18L:E4(5882'),E5(8792')

Vacate C3,C4 by ATC only. Max Taxi SPD 20KTS
 C2 HOLD SHORT 가까움(Vacate TaxiSPD)

RKPK(PUS) 13ft		RKSS(GMP) 59ft	
KE Gimhae 129.2 DCL -5분		PA KE GMP 131.15	
Rwy 32L Landing (06:00L~0900L / 12:00L~15:00L /18:00L~21:00L)			
PUS : NADP CLB 1500, 360000lbs MAX, CONF 1+F			
36	SOORO x KALOD tx	HD 002 ALT ATC Modi NADP	
18	GIMHAE x	HD 182 ALT 5000 Modi NADP	
KMH 113.8	PSN 114.0	36L 108.5	36R 109.5
FIX	36 : KMH R091, R271, R185		
RWY36 400ft Man L/H turn, Max Taxi SPD 20KTS			



Domestic

GMP : STAR			
ILS 32L/R	GUKDO xT	BUMSI	GUKDO 160
ILS 14R	GUKDO xU	DOKDO	GUKDO 160
FIX	KIP /8(RWY 32), YJU R271, P73 /2		
32L : D3(6532'), E2(9117'), 32R (No CL L/T) : E1(6614')			
14R : C1(6578')			
32L/R : 8 KIP L/G, 14R : LOC CAPT L/G			
FAF : Final Flap			
TWR -> GND -> APRON (All by ATC)			
Except RWY14R Landing (Until R)			

RKPC(CJU) 119ft	RKPK(PUS) 13ft
------------------------	-----------------------

KE CJU 129.4

DCL -10분

PA

KE Gimhae 129.2

CJU : SID (NADP 1)

07

AKPON xE

HD 066 ALT 9000

25

AKPON xW

HD 246 ALT ATC

YDM 109.0

07 109.9

25 111.3

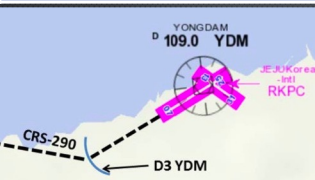
FIX

07 : NONE

25 : YDM246/3, R290

07 : Passing G4 CTC TWR

25 : 31 Holding PSN on P, E1,2,3 CTC TWR



Domestic

PUS : STAR (T/W 10tks, 36R, 360000lbs 제한)

ILS 36

KEVOX x

ANROD

9DME LG, 8DME FLAP

VOR 18

GAYHA x

ANROD

18 Circling Click!!

FIX

36 : IKMA/IKHE /9, /8

18 : KMH R284, R280

36L : C4 (6299'), C2(7795') / 36R : E3(5866'), E2(7339')

18R(DIS):C6(5770'),C7(6824') 18L:E4(5882'),E5(8792')

Vacate C3,C4 by ATC only. Max Taxi SPD 20KTS

C2 HOLD SHORT 가까움(Vacate TaxiSPD)

RKPK(PUS) 13ft**RKPC(CJU) 119ft**

KE Gimhae 129.2

DCL -5분

PA

KE CJU 129.4

PUS : NADP CLB 1500, 360000lbs MAX, CONF 1+F

36

**SOORO x
TOPAX tx**HD 002 **ALT ATC** **Modi NADP**

18

**BULIM x
ENGOT tx**HD 182 **ALT 5000** **Modi NADP****KMH 113.8****PSN 114.0****36L 108.5****36R 109.5**

FIX

36 : KMH R091, R271, R185

RWY36 400ft Man L/H turn, Max Taxi SPD 20KTS

**Domestic****CJU : STAR****AFT Merge PT(220kts) DCT IAF(210kts), FAF (160kts)****ILS Z 07****UPGOS xP****YUMIN****RNP Y 07 (No LPV), RNP Z 07 AR (RNP 0.11)****ILS Z 25****UPGOS xT(xM)****DUKAL**

FIX

RWxx /8

07 : P6(5176'), P5(5882'), P4(6840'-ATC HIRO)**25 : P7(5219'), P8(5882'), P10(7524'-ATC HIRO)**

Entering Rapid TWY CTC GND 121.675, STOP X

HST 40KTS

<u>RKSI(ICN) 23ft</u>	<u>RKPK(PUS) 13ft</u>
------------------------------	------------------------------

KE ICN 131.5 DCL -10분 TOBT 5분 차이시 CTC Comm	PA KE Gimhae 129.2
--	---------------------------

ICN : SID (33/34 NADP 1, 15/16 NADP 2)
--

33L/R	OSPOT xE/A	HD 333 ALT 5500/ATC SPD 230/x
34L/R	OSPOT xY	HD 333 ALT ATC SPD 230
15L/R	OSPOT xC	HD 153 ALT 5000 NADP2
16L/R	OSPOT xH	HD 153 ALT 5000 NADP2

NCN 113.8	33L 109.3	33R 108.9	15L 111.9	15R 109.1
WNG 112.9	34L 109.95	34R 108.1	16L 110.35	16R 108.55

FIX	33L/R : NC05L/R, R242 YJU R271	34L/R : EO34L/R, R242 YJU R271
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Parallel TWY 10KTS 이상(R17 MAX 15kts)

<u>Domestic</u>

PUS : STAR (T/W 10tkts, 36R, 360000lbs 제한)
--

ILS 36	KEVOX x	MASTA	9DME LG, 8DME FLAP
VOR 18	GAYHA x	MASTA	<u>18 Circling Click!!</u>
FIX	36 : IKMA/IKHE /9, /8		18 : KMH R284, R280

36L : C4 (6299'), C2(7795') / 36R : E3(5866'), E2(7339')
18R(DIS): C6(5770'),C7(6824') 18L:E4(5882'),E5(8792')

Vacate C3,C4 by ATC only, Max Taxi SPD 20KTS C2 HOLD SHORT 가까움(Vacate TaxiSPD)

RKPK(PUS) 13ft**RKSI(ICN) 23ft**

KE Gimhae 129.2

DCL -5분

PA

KE ICN 131.5

PUS : NADP CLB 1500, 360000lbs MAX, CONF 1+F**36****SOORO x**
KALOD tx**HD 002 ALT ATC**
Modi NADP**18****GIMHAE x****HD 182 ALT 5000 Modi NADP****KMH 113.8****PSN 114.0****36L 108.5****36R 109.5****FIX****36 : KMH R091, R271, R185****RWY 36 400ft Man L/H turn, Max Taxi SPD 20KTS****Domestic****ICN : STAR****ILS 33/34****GUKDO xE****ENPIL****GUKDO 180****ILS 15/16****GUKDO xH****MUNAN****GUKDO 180****FIX****RWY /8(180kts), /5(160kts) , YJU R271****33R : C4(7529'), C5(8513'), 33L : B4(7463'), B5(8513')****15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641')****34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507')****16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444')****8NM 180kts, 5NM 160kts, Parr TAXI 10kts이상, HIRO**

<u>RKSI(ICN) 23ft</u>	<u>RJBB(KIX) 17ft</u>
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KE ICN 131.5 DCL -10분 TOBT 5분 차이시 CTC Comm	PA	KE KIX 130.95
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ICN : SID (33/34 NADP 1, 15/16 NADP 2)
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33L/R	OSPOT xE/A	HD 333 ALT 5500/ATC SPD 230/x
34L/R	OSPOT xY	HD 333 ALT ATC SPD 230
15L/R	OSPOT xC	HD 153 ALT 5000 NADP2
16L/R	OSPOT xH	HD 153 ALT 5000 NADP2

NCN 113.8	33L 109.3	33R 108.9	15L 111.9	15R 109.1
WNG 112.9	34L 109.95	34R 108.1	16L 110.35	16R 108.55

FIX	33L/R : NC05L/R, R242 YJU R271	34L/R : EO34L/R, R242 YJU R271
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Parallel TWY 10KTS 이상(R17 MAX 15kts)

<u>DEP 125.15 – TGU 134.17 – FUK 124.15</u>	<u>Japan</u>
<u>TKO 133.8</u>	
<u>KIX RDR 120.85</u>	
<u>KIX APP 120.25</u>	

KIX : STAR (SAEKI 170, RANDY 150)

06L	ALISA B	BERRY	ILS Y 06L
06R	ALISA A	ALLAN	ILS Y 06R
24L/R	ALISA C	MAYAH	ILS Z 24L/R
FIX	RWxx /8		

06L : B8(5160'), B6(6751'), 24R : B7(5318'), B9(6751')
06R : A7(5137'), A6(6938'), 24L : A8(5269'), A9(6976')

RWY06 : After 2500ft L/G DN, After 1500ft L/D FLAP TAXI RTE 1(via J4), 2(via J3)

RJBB(KIX) 17ft		RKSI(ICN) 23ft		
KE KIX 130.95 DCL -15분		PA KE ICN 131.5		
KIX : SID – SOUJA tx (NADP 1)				
06L/R	HELEN x - SOUJA tx	HD 059 ATC (9000)		
24L/R		HD 239 ATC (9000) SPD 210		
KIE 111.6	06L 108.7	06R 108.1	24L 110.7	24R 108.5
APU Start, TAXI RTE 1(via J4), 2(via J3)				
DEP 119.2 TKO 132.7 – 133.8 FUK 124.15 IGU 120.57 APP 119.75 Japan				
ICN : STAR				
ILS 33/34	GUKDO xE	ENPIL	GUKDO 180	
ILS 15/16	GUKDO xH	MUNAN	GUKDO 180	
FIX	RWY /8(180kts), /5(160kts) , YJU R271			
33R : C4(7529'), C5(8513'), 33L : B4(7563'), B5(8513') 15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641')				
34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507') 16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444')				
8NM 180kts, 5NM 160kts, Parr TAXI 10kts이상, HIRO				

RKSI(ICN) 23ft	RJAA(NRT) 135ft
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KE ICN 131.5 DCL -10분 TOBT 5분 차이시 CTC Comm	PA KE Tokyo 131.70
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ICN : SID (33/34 NADP 1, 15/16 NADP 2)

33L/R	EGOBA xE/A		HD 333 ALT 5500/ATC SPD 230/x		
34L/R	EGOBA xY		HD 333 ALT ATC		
15L/R	EGOBA xC		HD 153 ALT 5000		
16L/R	EGOBA xH		HD 153 ALT 5000		
NCN 113.8		33L 109.3	33R 108.9	15L 111.9	15R 109.1
WNG 112.9		34L 109.95	34R 108.1	16L 110.35	16R 108.55
FIX	33L/R : NC05L/R, R242, YJU R271			34L/R : EO34L/R, R242, YJU R271	

Parallel TWY 10KTS 이상(R17 MAX 15kts)

[DEP 125.15 – TGU 134.17](#)
[TKO 124.15 – 132.02 – 124.1– 128.2](#)
[TKO APP 124.4 – 120.2](#)

Japan

NRT : HAKKA 330,YAGAN 240,LIVET 210,SWAMP 150

34L/R	SWAMP E (SWAMP T)	ELGAR (TYLER)	ILS 34L/R(Z)
16L/R	SWAMP G (SWAMP N)	GEMIN (NORMA)	ILS Z 16L/R
FIX	16L : ITM 4 / 34R : ITJ 14, 4 (DME) 16R : IKF 4 / 34L : IYQ 12, 4 (DME)		

16L : B6(6433'), B7(7017'), 34R : B4(5849'), B2(6778')
 16R : A6(6076'), A7(7624'), 34L : A5(6167'), A4(7641')

L/D DOWN before 14/12 DME, L/D FLAP 4 DME
 Arrival Taxi RTE in Jeppesen (No Numbering)

<u>RJAA(NRT) 135ft</u>	<u>RKSI(ICN) 23ft</u>
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KE Tokyo 131.70 DCL -15분	PA	KE ICN 131.5
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NRT : SID – ENPAR tx (NADP 1)

16L/R	TETRA x ENPAR tx	HD 157 ALT ATC		
34L/R		HD 337 ALT 7000/ATC		
NRE 117.9	16L 110.7	16R 111.5	34L 111.9	34R 110.9

APU Start, TAXI RTE 1, 2, 3, 4 RWY 별 DEP RTE

DEP 124.2

TKO 120.5 – 133.45 – 133.02 – 133.8

TGU 120.57

APP 119.75

Japan

ICN : STAR

ILS 33/34	GUKDO xE	ENPIL	GUKDO 180
ILS 15/16	GUKDO xH	MUNAN	GUKDO 180
FIX	RWY /8(180kts), /5(160kts) , YJU R271		

33R : C4(7529'), C5(8513'), 33L : B4(7563'), B5(8513')
15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641')
34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507')
16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444')

8NM 180kts, 5NM 160kts, Parr TAXI 10kts이상, HIRO

RKSI(ICN) 23ft	RJTT(HND) 21ft
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KE ICN 131.5 DCL -10분 TOBT 5분 차이시 CTC Comm	PA Delta Oper 132.075
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ICN : SID (33/34 NADP 1, 15/16 NADP 2)

33L/R	EGOBA xE/A		HD 333 ALT 5500/ATC SPD 230/x		
34L/R	EGOBA xY		HD 333 ALT ATC		
15L/R	EGOBA xC		HD 153 ALT 5000		
16L/R	EGOBA xH		HD 153 ALT 5000		
NCN 113.8		33L 109.3	33R 108.9	15L 111.9	15R 109.1
WNG 112.9		34L 109.95	34R 108.1	16L 110.35	16R 108.55
FIX	33L/R : NC05L/R, R242, YJU R271			34L/R : EO34L/R, R242, YJU R271	

Parallel TWY 10KTS 이상(R17 MAX 15kts)

DEP 125.15 – TGU 134.17 – FUK 133.02	Japan
TKO 120.5 – TKO 133.35	
TKO APP 119.1 – 119.65	

HND : **STAR XAC Night– APP xxx Y 1400z~ SPENS 220**
22,23 LDA Taiload Procedure

34L/R	XAC xK/H	KAIHO/CACAO	ILS X / VIS
22	XAC xB	BACON	LDA W(RNV22-W)
16R/L	XAC R	NATTY/SANDY	RNP(RNV16R/L)
23	-	DANON	LDA W(RNV23-W)
FIX	Z 34L : IHA /10, /5		Z 34R : ITC /10, /5

34L:L12(6515),L13(7165), 16R(**DIS**):L5(5147),L3(6361)
22 : B4(6207), B3(6830), 23: D5(5072),D3(6391)
34R(**DIS**):C10(5780),C11(7270)
16L(**DIS**):C6(5925),C4(7004)

xxx Z : 180kts, 160kts limit APP Chart, xxx Y After 1400z
Gate 105P Procedure

RJTT(HND) 21ft**RKSI(ICN) 23ft**

Delta Oper 132.075

DCL -15분

PA

KE ICN 131.5

HND : SID (xx B/C 2200-0230z 0600-1000z) NADP 1

ALL

BEKLA x
OPPAR x

RWY HD(348(158),231(051))

ALT ATC

Mod SPD 230

HME
112.234L
111.716R
111.5534R
108.916L
111.9522
108.123
110.5

FIX

34L : HME 351/1.1, R095, 34R : HME R080,
R095, 22 : HME /2.2 R18534R BEKLA : KAIJI 230kts, TO FLAP SPD Until
TORAM

05 OPPAR : Reduce SPD ARC TURN

RWY05 RTE5 TAXI Chart

DEP ATISTKO 120.5 – FUK 133.02TGU 120.57APP 119.75**Japan**

ICN : STAR

ILS 33/34

GUKDO xE

ENPIL

GUKDO 180

ILS 15/16

GUKDO xH

MUNAN

GUKDO 180

FIX

RWY /8(180kts), /5(160kts) , YJU R271

33R : C4(7529'), C5(8513'), 33L : B4(7563'), B5(8513')

15L : C2(7522'), C1(8536'), 15R : B3(7454'), B2(8641')

34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507')

16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444')

8NM 180kts, 5NM 160kts, Parr TAXI 10kts이상, **HIRO**

CRZ FUEL Penalty (Approximation)

A330

4000ft below OPT ALT : 5% increase trip fuel
8000ft below OPT ALT : 10% increase trip fuel
Above OPT ALT : 3% increase trip fuel

FUEL Consumption

APU

GND : 438 LBS/hr (BLOCK FUEL산정)
IN FLT : 286 LBS/hr

TAXI

2 ENG, no APU : 3300 LBS/hr (55 LBS/hr)

CRZ

- 100ft 당 8 min
ex) 1000ft 상승 80 min, 1hr 20min

Holding

분당 200LBS (Actual F/F 참조)

Missed App & Landing

3000 LBS (과거 EDTO자료 1회 Missed App + L/D)

FUEL Loading

Tail Tank :

CL TANK 80000LBS 이상 5000LBS 까지

CL TANK 160000LBS 이상 10000 LBS 까지

FUEL Overfill : 1000LBS 기준

PACKS USE WITH LP AIR CONDITION UNIT

공항 요구로 APU OFF후 기내 온도 조절을 위한 방법
Low Press Unit을 연결하며 APU와 동시사용 금지

**The flight crew must not use conditioned air
from the packs and from the LP Air**

**Conditioning Unit at the same time, to prevent
any adverse effect on the Air Conditioning
system.**

절차 POM, FCOM에 없음

APU BLEED USE WITH HP AIR START UNIT

APU 부작동시 AIR START UNIT으로 시동을 위해 사용
외부 BLEED AIR의 역할을 함. APU BLEED 동시사용
금지

**The flight crew must not use bleed air from
the APU BLEED and from the HP Air Start Unit
at the same time, to prevent any adverse
effect on the Bleed Air System.**

절차 POM, FCOM에 없음

STARTING with AIR START UNIT

통상 ENG 1 먼저 (EXT PWR Start 2 ENG)
“Req Engine Start up Present Positon~~~”

- Before connecting the air start unit:

PACK 1 & 2 OFF F/O

prevent any possible contamination of the packs by the air start unit.

APU BLEED OFF F/O

ENG 1 & 2 BLEED OFF F/O

X BLEED OPEN F/O

AIR START UNIT CONNECT ... REQ CAP

Two ground air start units may be used in parallel if the pressure/flow relation is expected to be marginal.

- When cleared to start:

Note: The **AIR ABNORM BLEED CONFIG** alert is triggered after the first engine start. It can be disregarded.

ENG 1 START CAP

Note: As necessary, engine 2 can be started first.

In this case, check the brake ACCU pressure prior to engine start.

Note: The minimum recommended starter air supply pressure is 25 PSI when the start valve is open.

Home

- After engine 1 is started:

- If the air start unit is used to start engine 2:

ENG 2 START CAP

- When engine 2 is started:

AIR START UNIT DISC REQ CAP

X BLEED AUTO F/O

ENG 1 & 2 BLEED ON F/O

PACK 1 & 2 ON F/O

- If the crossbleed engine start procedure is used to start engine 2:

AIR START UNIT DISC REQ CAP

ENG 1 BLEED ON F/O

PACK 1 & 2 ON F/O

CRSBLD ENG START PROC.APPLY

EXT PWR CHK AVAIL CAP

EXT PWR DISCONNECT REQ CAP

FCOM : In order to ensure that the entire network remains electrically supplied, it is recommended to start both engines before the disconnection of the external electrical power.

NOR PROC - AFTER STARTRESUME

ENG CROSSBLEED START

#1 ENGBLEED 로 #2 ENG START, EXT PWR
필요

**PushBack 완료(Parking Brake set), Supply eng
Area CLR**

One engine must be running in order to supply air
for other engine start.

● Before second engine start :

APU BLEED OFF F/O

The BLEED valve of the supplying engine reopens
and the cross bleed valve closes.

ENG BLEED (supply engine) ON F/O

ENG BLEED (receive engine) OFF F/O

The bleed valves of receiving engines are closed to
avoid reverse flow leakage.

X BLEED..... OPEN F/O

● When cleared to start :

AREA CLEAR OF OBS CONF CAP

Ensure increased power jet wake does not constitute
any hazard to people or installations behind the
aircraft.

THR LVR (sup eng) . ADJ BLD PRES CAP

Adjust thrust of supplying engine to obtain an
engine bleed pressure of **30 PSI**. If the thrust
required to obtain the appropriate engine bleed
pressure exceeds **40 % N1**, pay particular attention
to the surrounding area.

RECEIVING ENGINE START ALL

Apply the normal engine start procedure.

● After start :

THR LVR (supply engine) IDLE CAP

X BLEED..... AUTO F/O

ENG BLEED (receive engine) ON F/O

PACK 1 & 2 ON F/O

COLD TEMP CORRECTION General

5도 간격은 보수적으로 보간법 적용됨

Min 제외한 모든 고도 수정은 ATC 인가 필요

Mandatory, Missed App 고도 ATC 사전 인가 없이 금지

반드시 고도 - FE 후의 고도를 보정해야함.

Ex) FE 200ft 공항 : 5000ft는 4800ft만 보정해야함.

Height Above FE (Feet) 200-800ft

TEMP	200	300	400	500	600	700	800
0	20	20	30	30	40	40	50
-5	20	30	40	40	50	60	70
-10	20	30	40	50	60	70	80
-15	30	40	50	60	80	90	100
-20	30	50	60	70	90	100	120

Height Above FE (Feet) 900-5000ft

TEMP	900	1000	1500	2000	3000	4000	5000
0	50	60	90	120	170	230	280
-5	70	80	120	160	230	310	390
-10	90	100	150	200	290	390	490
-15	110	120	180	240	360	480	600
-20	130	140	210	280	420	570	710

Domestic

Japan

China

GMP, CJU, PUS next page

Home

COLD TEMP CORRECTION (Domestic)

Min 은 반드시 수정 (중간 고도 CORRECTION은 PIC 결정)
Missed App 고도는 ATC 협조 필요

GMP 32L (261') / 32R (262') / 14R (254')

32L/R	8000	5500	5300	4000	2800	2300	2000
0	8450	5810	5600	4230	2970	2440	2120
-5	8620	5930	5710	4310	3030	2490	2160
-10	8780	6040	5820	4390	3080	2530	2200
R14	4000	2800	1400		4000		
0	4230	2970	1490		4230		
-5	4310	3030	1520		4310		
-10	4390	3080	1540		4390		

CJU 07 (307') / 25 (296')

	4000	2900	1800	07	8000	25	6000
0	4220	3070	1900		8450		6340
-5	4300	3130	1940		8620		6460
-10	4380	3180	1970		8780		6590

PUS 36L(233'),36R(228') / 18L/R (see below)

36L/R	6000	5000	3300	2100		6000	
0	6340	5290	3490	2210		6340	
-5	6460	5390	3560	2250		6460	
-10	6580	5490	3620	2290		6580	
18L/R	6000	5000	4000	2600	1700		6000
0	6340	5290	4230	2760	1800		6340
-5	6460	5390	4310	2810	1830		6460
-10	6580	5490	4390	2860	1870		6580

COLD Wx Operation

OAT (GND) / TAT (TAT) is 10°C (50°F) or below :

- visible moisture (clouds, fog with VIS 1SM (1600 m) or rain, snow, sleet, ice crystals...)
- ice, snow, slush and standing water is present on the ramps, taxiways, or runways.

PRELIMINARY (OAT 3°C 이하)

LOGBOOK ----- CHECK
MAX TIME TO NEXT ENG ACC --- DETERMINE
TAXI IN TIME – NEXT ENG ACC TIME 15분 이내인지 확인

TAXI-OUT(OAT 3°C 이하)

TAXI TIME ----- MONITOR
15분 초과 또는 Eng Vibrations시 절차 수행

SOP SURFACE COND & AREA ----- CHECK
ATC ----- NOTIFY

PARKING BRAKE ----- ON
the aircraft starts to move, immediately retard the thrust levers to IDLE.

THR LEVERS ----- 50% N1 (NO HOLD TIME)

THR LEVERS ----- RETARD TO IDLE

Repeat this procedure at intervals not longer than 15 min, or if the engine vibrations increase.

SOP TAXI ----- RESUME

BEFORE TAKEOFF

FLAPS lever ----- SET FOR TAKEOFF

FLAPS ----- CHECK PSN

T/O CONFIG pb ----- TEST

T/O MEMO ----- CHECK NO BLUE

AFTER START checklist ----- COMPLETE

BEFORE TAKEOFF checklist --- ACCOMPLISH

TAKEOFF(OAT 3°C 이하)

A final engine acceleration must be performed before takeoff to entirely clean the engine from ice accretion.

ATC ----- NOTIFY

PARKING BRK(OR BRK PEDALS) ----- APPLY

THR LEVERS ----- 50% N1 (NO HOLD TIME)

THR LEVERS ----- RETARD TO IDLE

SOP TAKEOFF ----- RESUME

AFTER L/D, PARKING

DO NOT RETRACT FLAPS

Ice Shedding 절차 반복

Home

ENG ON Deicing in ICN

TOBT- 40min CTC KE ICN (사전신청, 결과확인)

ICN Deicing "Deicing Required ENG On Deicing"

ICN Apron "Req Pushback Deicing Zone xx" SQ2000

Pad Control Arrange Deicing Pad No.

Ice Man Manage Deicing Process

Cold Wx

Gate 시동후 (FLAP UP, FLT CTL CHK 없이)

PARKING BRAKE ----- SET

Report Parking Brake SET - > Ice Man

BROADBAND XMT sw ----- OFF

COMMUNICATION with GND ----- ESTAB

DE/ANTI-ICING FLUID TYPE ----- CHECK

CABIN PRESS MODE SEL ---- CHK AUTO

ENG 1 & 2 BLEED ----- OFF

Note : AIR ENG 1+2 BLEED FAULT – Disregard

APU BLEED ----- OFF

DITCHING pb ----- ON

Note : CAB PR FWD OFV NOT OPEN, CAB PR AFT OFV NOT OPEN, COND LAV + GAL VENT FALUT –

Disregard

THRUST LEVERS ----- CHK IDLE

"A/C PREPARED FOR SPRAYING"

Report Ready for Deicing - > Ice Man

START SPRAY REQ DCL(CTC DEL)

UPON COMPLETE SPRAY(TIME CHK 1분)

용액과 마지막 용액 뿌린 시간 받고 적는다.

Holdover Time 결정!!!

PITOT STATICS (GND Crew) ----- CHK

GND EQUIPMENT ----- REMOVE

DITCHING pb ----- OFF

OUTFLOW V/V - CHK OPEN(ECAM press)

TIME CHECK 1분후

ENG 1 & 2 BLEED ----- ON

BROADBAND XMT sw ----- ON

5분후

APU BLEED ----- AS RQRD

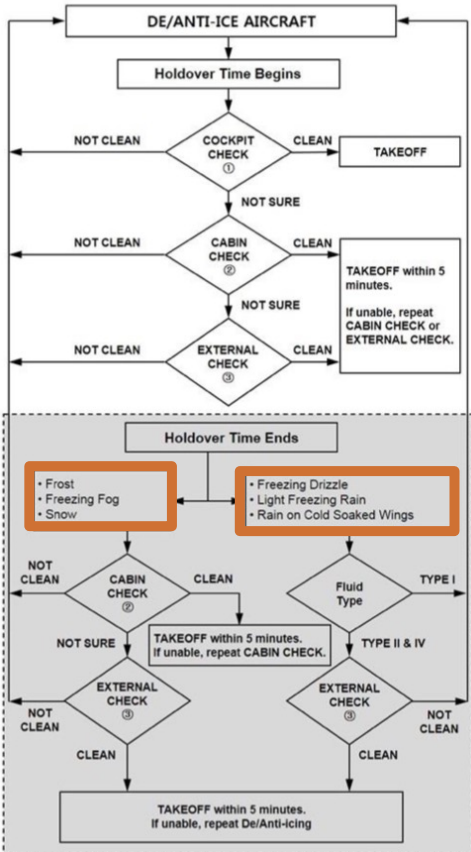
NORMAL PROCEDURE ----- RESUME

FLT CONTROL CHK, FLAPS UP

DECISION TREE next page

Home

TAKEOFF DECISION TREE



ENG OFF Deicing in GMP...

TOBT- 20min CTC KE GMP (PAD, New TOBT)

REQ DCL

Deicing "Deicing Required PADxxx" ± 5 min TOBT
Apron "Req Pushback Deicing PADxxx"

Gate 시동후 (FLAP UP, FLT CTL CHK, APU OFF 없이)

Engine anti-ice sw ----- OFF

PARKING BRAKE ----- SET

BROADBAND XMT sw ----- OFF

COMMUNICATION with GND ----- ESTAB

DE/ANTI-ICING FLUID TYPE ----- CHECK

CABIN PRESS MODE SEL ---- CHK AUTO

(HL7524-7720) GND COOL ----- OFF

ENG 1 & 2 BLEED ----- OFF

APU BLEED ----- OFF

DITCHING pb ----- ON

Note : CAB PR FWD OFV NOT OPEN, CAB PR AFT OFV NOT
OPEN, COND LAV + GAL VENT FALUT -Disregard

ENG MASTER sw ----- OFF

SHUTDOWN CHECKLIST

"A/C PREPARED FOR SPRAYING"

UPON COMPLETE SPRAY (TIME CHK 1분)

용액과 마지막 용액 뿌린 시간 받고 적는다.

Holdover Time 결정!!!

(HL7524-7720) GND COOL ----- OFF

PITOT STATICS (GND Crew) ----- CHK

GND EQUIPMENT ----- REMOVE

DITCHING pb ----- OFF

OUTFLOW V/V - CHK OPEN (ECAM press)

TIME CHECK 1분후

ENG 1 & 2 BLEED ----- ON

BROADBAND XMT sw ----- ON

5분후

APU BLEED ----- ON

NORMAL PROCEDURE ----- RESUME

Note : INIT B check & re-enter

PREFLT CHKlist -> Req STARTUP -> CHKlist

BOTH ENGINES ARE STARTED

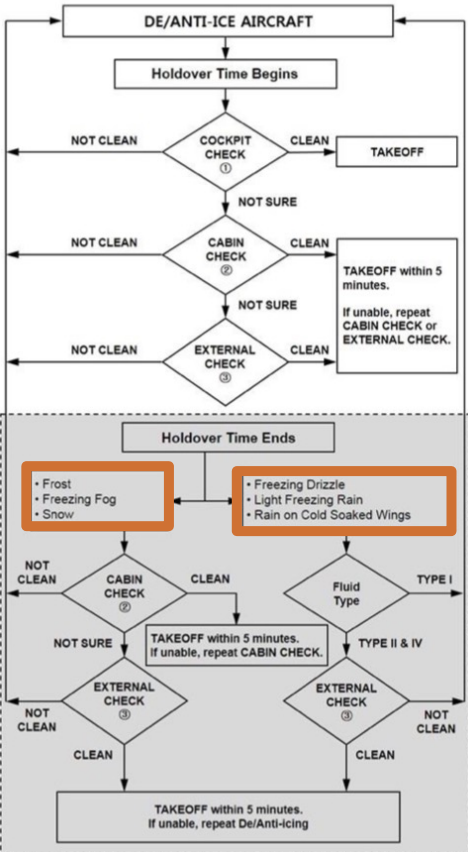
FLT CTL CHK, FLAPS UP, APU as REQ

DECISION TREE next page

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[Cold Wx](#)

TAKEOFF DECISION TREE



PUS VOR 18L/R

F-PLN			SEC F-PLN	
APPR	VIA	STAR	APPR	VIA
VOR18L/R	GAYHA	GAYHA x TRANS MASTA	R18L/R	V18L/R (MDA: Blank) (LDG CONF chk)

FIX : KMH 280(Base Turn), 284(Missed App)



Missed App

Base Turn 이전 : L/H Turn **KMH 284** OUTBD (PULL TRK)

Base Turn 이후 : Continue R/H Turn **KMH 284** OUTBD (PULL TRK)

PUS LOC 36L/R Circling 18L/R

F-PLN

APPR	VIA	STAR
LOC36L	KEVOX	KEVOX x TRANS MASTA

SEC F-PLN

APPR	VIA
R18L/R	V18L/R (MDA: Blank) (LDG CONF chk)

CI36L(CF36R) 3500 FI36L(FF36R) 2100

FIX : KMH 280(Base Turn), 310(Missed App)



Missed App

Base Turn 이전 : L/H Turn **KMH 310** OUTBD (PULL TRK)

Base Turn 이후 : Continue R/H Turn **KMH 310** OUTBD (PULL TRK)

Domestic

GS KTS	KM	MILES
300	560	350
310	570	360
320	590	370
330	610	380
340	630	390
350	650	400
360	670	410
370	690	430
380	710	440
390	720	450
400	740	460
410	760	470
420	780	480
430	800	500
440	820	510
450	830	520
460	850	530
470	870	540
480	890	550
490	910	560
500	930	580
510	950	590
520	960	600
530	980	610
540	1000	620
550	1020	630
560	1040	650
570	1060	660
580	1070	670
590	1090	680
600	1110	690
610	1130	700
620	1150	710
630	1170	730
640	1190	740
650	1200	750
660	1220	760
670	1240	770
680	1260	780
690	1280	800
700	1300	810